

Anshul Raje

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U.S. Citizen (No sponsorship required)

EDUCATION

Worcester Polytechnic Institute

M.Sc. Robotics Engineering

Related Coursework: Deep Learning for Perception, Hands-on Autonomous Aerial Robotics

Aug 2026 — Expected May 2028

Worcester, MA

BITS Pilani, Goa

B.E. Electronics and Communication Engineering — GPA: 3.401/4.0

Sept 2020 – Jul 2024

Goa, India

SKILLS

Languages:	Python, C/C++, YAML, XML
Perception & Autonomy:	Computer Vision, Object Detection, YOLO, Synthetic Dataset Generation, Localization, Robot Modeling, Simulation-to-Real, Behavior Trees, Motion Planning, Path Planning
Simulation & Tools:	OpenCV, Ultralytics, ROS 2, ROS, ROS 2 Control, ROS 2 Lifecycle Management, Gazebo (robot simulation), RViz (visualization), Docker, Git, Linux
Controls & Estimation:	State Estimation, Kalman Filter, IMU, MPC, Foothold Optimization, Actuator Characterization, Stanley Controller, Pure Pursuit, Forward & Inverse Kinematics
Embedded & Hardware:	Depth Cameras (Stereolabs ZED, Intel RealSense), NVIDIA Jetson (Orin/Xavier/Nano)

EXPERIENCE

Strider Robotics

Associate Robotics Engineer

Aug 2024 – Apr 2026

Bangalore, India

- Computer Vision Pipelines:** Developed ROS 2 nodes for analog gauge reading and fire extinguisher detection using YOLO models trained on custom datasets via Ultralytics; deployed and optimized inference on NVIDIA Jetson hardware onboard the robot, integrated as independently testable lifecycle nodes.
- Perception Orchestration Framework:** Designed and built a 5-layer modular ROS 2 architecture for industrial inspection robots, separating hardware drivers, CV application nodes, and task orchestration — deployed on quadruped robots for client's field operations.
- Config-Driven Task Execution:** Architected a behavior tree-based orchestrator with ROS 2 lifecycle management, enabling perception tasks to be implemented in roughly 200 lines of code (80% YAML); deployed across multiple clients.
- State Estimation:** Designed a Kalman filter-based state estimator fusing IMU, joint encoder, and foot contact data to enable reliable localization in GPS-denied environments.
- Robust Locomotion:** Enhanced quadruped reliability by integrating MPC-based control with foothold optimization for stable traversal on uneven terrain; developed fall recovery strategies for autonomous self-righting.
- Actuator Characterization:** Built automated testing pipelines to identify actuator dynamics such as friction and inertia, reducing simulation-to-real discrepancies and accelerating reinforcement learning experiments.

HiPeRT Lab, University of Modena and Reggio Emilia

Research Intern — Robotics and Intelligent Systems

Jul 2023 – Dec 2023

Modena, Italy

- Computer Vision:** Developed a novel algorithm for synthetic dataset generation; trained YOLOv4 on a synthetic dataset of 1500+ images, achieving 97% detection accuracy for opponent vehicle localization.
- Motion Planning & Control:** Integrated an MPC-curve controller into the autonomous race car, achieving a mean lateral error of under 2 cm, replacing the underperforming Pure Pursuit controller.

PROJECTS

Project Kratos — Mars Rover Team | Controls Lead

Autonomous terrain traversal, life-sign detection, and robotic manipulation

Sept 2021 – May 2023

BITS Pilani, Goa

- Perception & Path Following:** Implemented Stanley Controller for waypoint tracking within 5 cm accuracy; integrated YOLOv3 trained on a custom dataset for real-time obstacle detection.
- Manipulator:** Developed Forward and Inverse Kinematics algorithms to precisely control a 5-DOF on-rover manipulator.

ACHIEVEMENTS

- 2nd Place Globally, Anatolian Rover Challenge (Jul 2022):** Placed 2nd internationally in Istanbul, Turkey; won Best Autonomous and Controls Vertical and Best Science Vertical.
- Excellence Award, International Rover Challenge (Jan 2023):** Received the Excellence Award for Overall Performance at the International Rover Challenge, Bangalore, India.